

# [EECS 304] Control I

## Sheet (6) Block Diagram Reduction

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# Outline

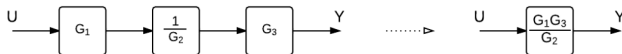
## 1 Overview

## 2 Problems

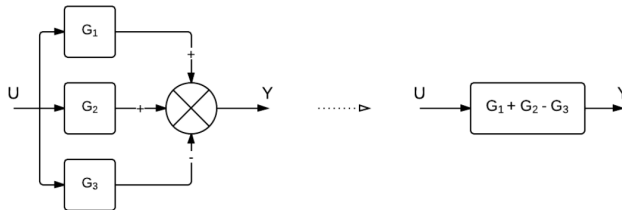
- Problem 1
- Problem 2
- Problem 3

# Overview

## First Rule: Cascaded Blocks

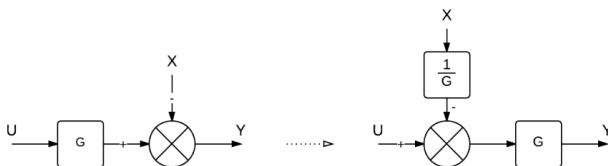


## Second Rule: Parallel Branches

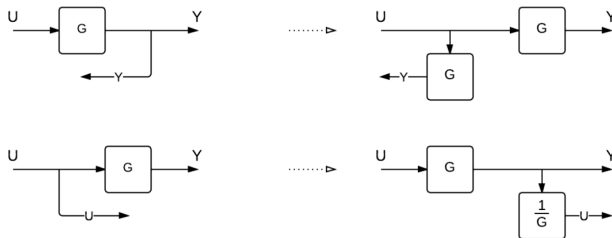


# Overview

## Third Rule: Shifting a summer behind of a block

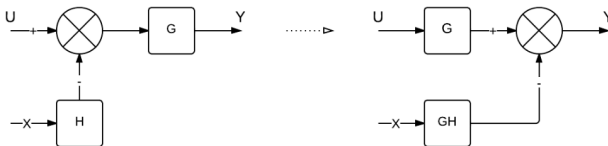


## Fourth Rule: Shifting take-off point

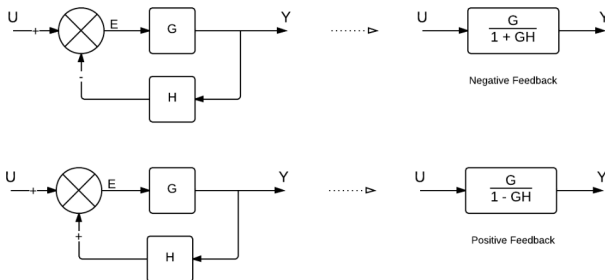


# Overview

## Fifth Rule: Shifting a summer in front of a block



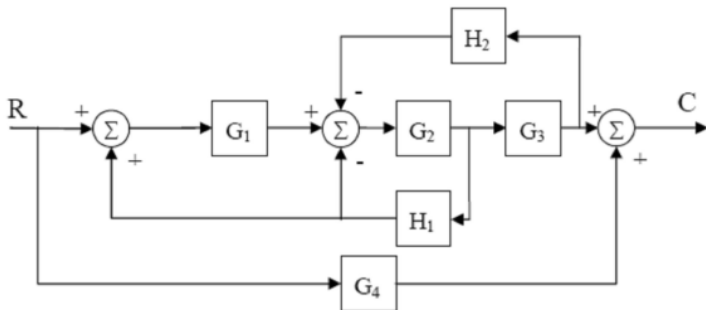
## Sixth Rule: Reduction of feedback



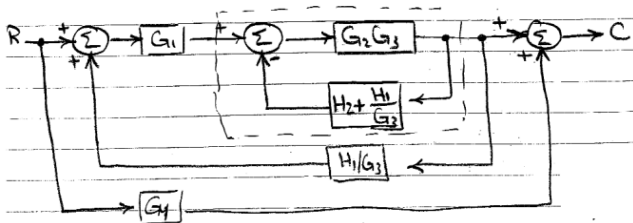
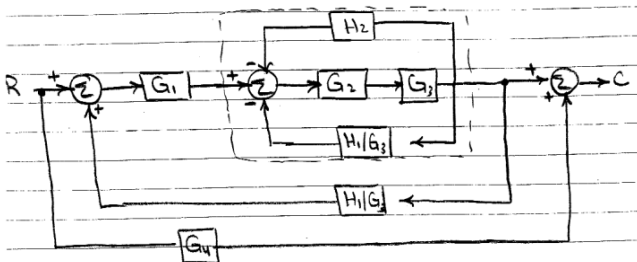
# Problem 1

## Problem 1

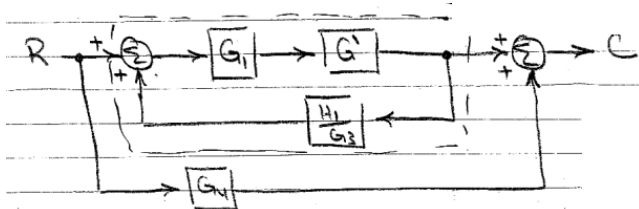
Reduce the following block diagram.



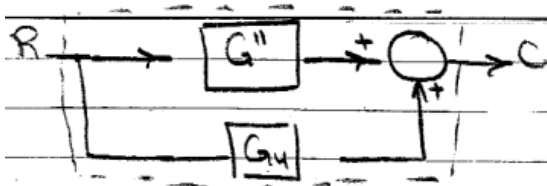
# Problem 1



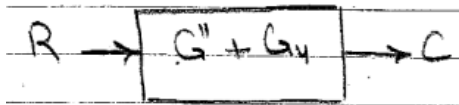
# Problem 1



$$G' = \frac{G_2 G_3}{1 + G_2 G_3 H_2 + G_2 H_1}$$



# Problem 1



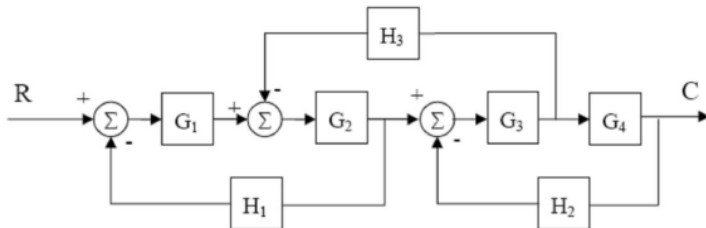
$$G'' = \frac{G_1 G_2 G_3}{1 + G_2 G_3 H_2 + G_2 H_1 - G_1 G_2 H_1}$$

$$\frac{C}{R} = G_4 + \frac{G_1 G_2 G_3}{1 + G_2 G_3 H_2 + G_2 H_1 - G_1 G_2 H_1}$$

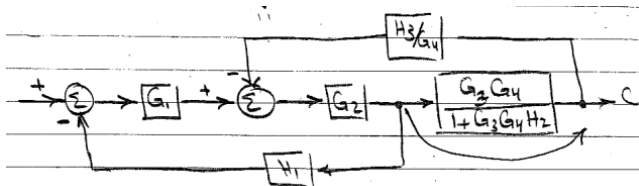
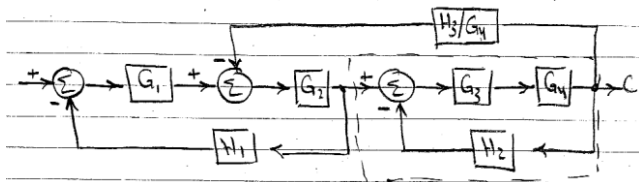
## Problem 2

### Problem 2

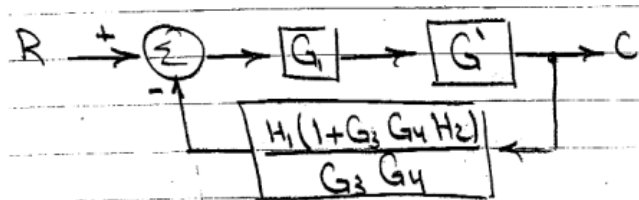
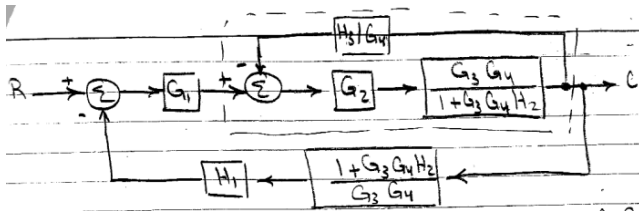
Determine  $C/R$  for the following system using block diagram reduction.



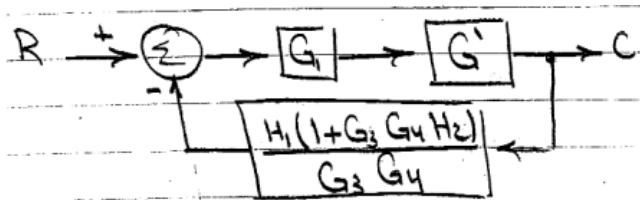
## Problem 2



## Problem 2



## Problem 2



$$G' = \frac{G_2 G_3 G_4}{1 + G_3 G_4 H_2 + G_2 G_3 H_3}$$

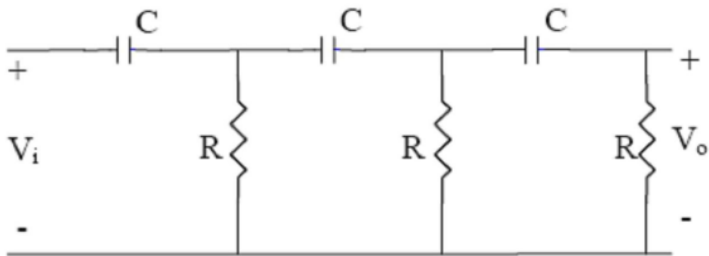
$$\frac{C}{R} = \frac{G_1 G'}{1 + G_1 G' \cdot \frac{H_1(1 + G_3 G_4 H_2)}{G_3 G_4}}$$

$$\frac{C}{R} = \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2 + G_2 G_3 H_3 + G_1 G_2 H_1 + G_1 G_2 H_1 G_3 G_4 H_2}$$

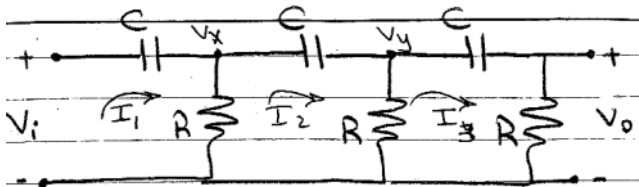
## Problem 3

### Problem 3

For the shown circuit, determine  $V_o/V_i$  using block diagram reduction method.



## Problem 3



$$I_1 = (V_i - V_x)sC$$

$$V_x = (I_1 - I_2)R$$

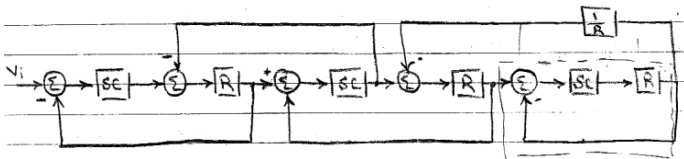
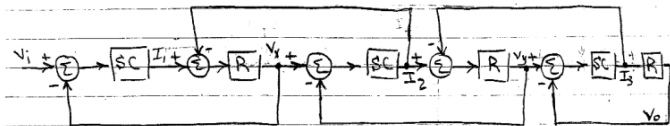
$$I_2 = (V_x - V_y)sC$$

$$V_y = (I_2 - I_3)R$$

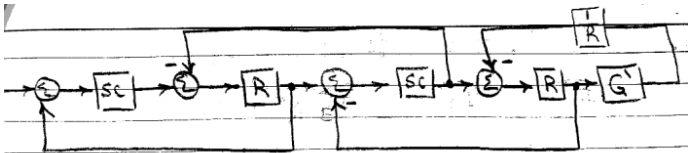
$$I_3 = (V_y - V_o)sC$$

$$V_o = I_3 R$$

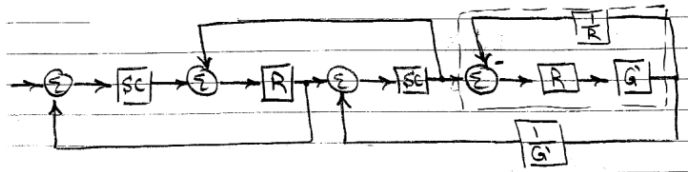
# Problem 3



## Problem 3

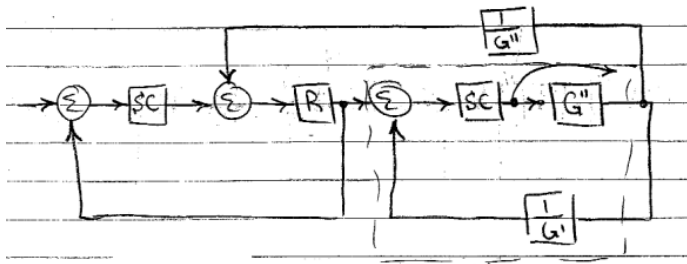


$$G' = \frac{sCR}{1 + sCR}$$



$$G'' = \frac{RG'}{1 + \frac{RG'}{R}} = \frac{sCR^2}{1 + sCR + sCR} = \frac{sCR^2}{1 + 2sCR}$$

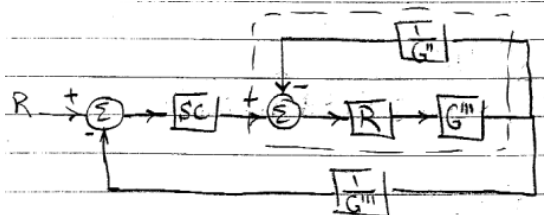
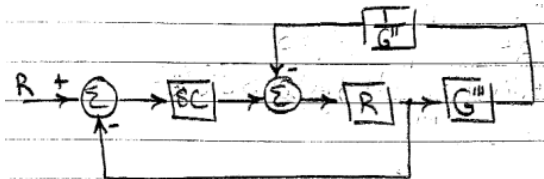
# Problem 3



$$G''' = \frac{G'' sC}{1 + \frac{sCG''}{G'}} = \frac{G' G'' sC}{G' + sCG''}$$

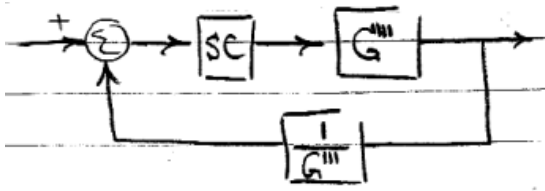
$$G''' = \frac{s^2 C^2 R^2}{1 + 3sCR + s^2 C^2 R^2}$$

# Problem 3



$$G'''' = \frac{RG''' G''}{G'' + RG'''} = \frac{s^2 C^2 R^3}{1 + 4sCR + 3s^2 C^2 R^2}$$

## Problem 3



$$TF = \frac{sCG''''G'''}{G'''+sCG''''} = \frac{s^3C^3R^3}{1+5sCR+6s^2C^2R^2+s^3C^3R^3}$$

# Thank you!

## Questions?

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